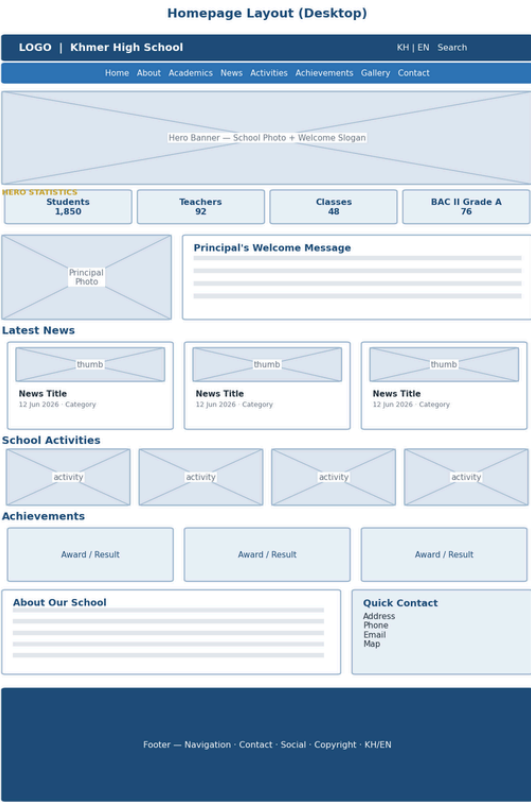


High School Website Management System

Bilingual (Khmer & English) Public Website + Content Management Platform



Software Requirements Specification & MVP Scope Document

Prepared for Cambodian High School Stakeholders

Version 1.0 · June 2026 · Status: Approved for Development

Document Control

This Software Requirements Specification (SRS) defines the business, functional, technical, and user-experience requirements for the High School Website Management System. It is the authoritative reference for the development team, project managers, UI/UX designers, QA engineers, and school stakeholders throughout the project lifecycle.

Document Information

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Project Name	High School Website Management System
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Revision History

Version	Date	Author	Summary of Changes
0.1	Jun 2026	BA Team	Initial draft from stakeholder requirement document
0.5	Jun 2026	Architecture	Added database, API, and architecture sections
1.0	Jun 2026	Project Lead	Baseline approved for MVP development

Approval

Role	Name	Signature	Date
School Director (Sponsor)			
Project Manager			
Solution Architect			
QA Lead			

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1. Executive Summary

The High School Website Management System is a modern, bilingual (Khmer and English) web platform that enables a Cambodian high school to communicate effectively with its students, parents, teachers, alumni, and the wider public. The platform combines a fast, mobile-first public website with an easy-to-use administrative back office that allows non-technical staff — most importantly the School Director — to keep school information, statistics, news, and achievements continuously up to date.

Many Cambodian schools either lack a web presence entirely or rely on social media pages that are difficult to organize, search, and trust as an official source. This project delivers an official, credible online identity for the school while removing the technical barriers that normally prevent school staff from maintaining a website. Every content type that changes regularly — enrollment dates, exam notices, activity photos, and headline statistics such as the number of BAC II Grade A students — can be updated through simple forms without any developer involvement.

1.1 Vision Statement

To give the school a trustworthy, bilingual digital front door that showcases its achievements, simplifies access to information for parents and students, and can be fully maintained by the school's own staff.

1.2 Key Outcomes

- A professional public website that loads in under three seconds on mobile networks and works fully in both Khmer and English.
- A self-service statistics dashboard that lets the Director update headline numbers in under one minute, with changes appearing on the homepage immediately.
- Centralized management of news, announcements, activities, achievements, media galleries, and downloadable documents.
- Role-based administration with audit logging, so the school controls exactly who can publish what.
- Strong SEO and social-sharing support so the school is discoverable and presents well when shared.

1.3 Scope at a Glance

Area	MVP (Phase 1)	Future (Phase 2)
Public website	8 core modules + 4 recommended features	Multi-campus, blog comments
Administration	CMS, statistics, users & roles, audit logs	Workflow approvals, analytics suite
Languages	Khmer + English	Additional languages
Integrations	Google Maps, email notifications	SMS, Telegram, payment, e-library
Access	Public read + staff admin	Parent/student logins, portal

The remainder of this document expands each of these areas into business requirements, functional requirements, user stories, non-functional requirements, database and API designs, UI/UX specifications, a development roadmap, and a clearly bounded MVP scope.

2. Business Requirements

2.1 Background & Problem Statement

The school currently lacks a single, authoritative online channel through which it can publish official information. Parents seeking enrollment requirements, students looking for exam schedules, and alumni or NGOs wanting to verify the school's standing have no reliable destination. Information is scattered across informal social-media posts that are hard to search, easily lost in feeds, and not perceived as official.

At the same time, school leadership does not have technical staff. Any solution that requires editing code or contacting a developer for routine updates will not be maintained and will quickly become stale. The core business problem is therefore twofold: the school needs a credible public presence, and it needs to be able to maintain that presence entirely on its own.

2.2 Business Objectives

#	Objective	Description
BO-1	Establish official online presence	Provide a single trusted source for school information accessible to all stakeholders.
BO-2	Enable non-technical content management	Allow the Director and staff to update statistics, news, and activities without developer help.
BO-3	Improve parent & student access to information	Make enrollment, exam, and calendar information easy to find in Khmer and English.
BO-4	Showcase school performance & achievements	Highlight headline statistics and awards to build reputation and attract enrollment.
BO-5	Reduce administrative workload	Offer downloadable forms and self-service information to cut repetitive enquiries.
BO-6	Ensure long-term sustainability	Keep hosting and maintenance affordable and operable by school staff.

2.3 Success Metrics (KPIs)

The project's success will be measured against the following indicators within the first three months after launch:

KPI	Target
Homepage load time (mobile, 4G)	Under 3 seconds
Time for Director to update a statistic	Under 1 minute, no developer involvement
Content updates published by school staff	100% self-service (zero developer tickets for content)
Bilingual coverage of public pages	100% of public content available in Khmer and English
Monthly unique visitors (3 months post-launch)	Baseline established and trending upward
Contact-form enquiries answered	Email notification delivered to admin within 1 minute

2.4 Stakeholders

Stakeholder	Interest / Role	Type
School Director	Primary sponsor; updates statistics; approves content	Internal / Admin
Vice Directors	Review and contribute content; oversee departments	Internal / Admin
Administrative Staff	Day-to-day content entry (news, activities, media)	Internal / Editor
Website Administrator	Manages users, permissions, configuration, integrity	Internal / Admin
Students	Read news, exam info, activities, achievements	External / Public
Parents	Read enrollment info, notices, calendar, contact	External / Public
Alumni & Community	Follow achievements, activities, history	External / Public
NGOs & Gov. Officers	Verify school information and performance	External / Public

2.5 Scope

2.5.1 In Scope (MVP)

- Public bilingual website with the eight core modules and four recommended features defined in Section 4.
- Administrative back office for content, statistics, media, users, roles, and audit logs.
- Google Maps embedding and email notifications for contact messages.
- SEO, Open Graph, sitemap generation, and mobile-responsive design.

2.5.2 Out of Scope (MVP)

- Student or parent login accounts, grade portals, or fee payment.
- E-learning, online classes, or library management.
- SMS / Telegram integration and automated content-approval workflows (planned for Phase 2).
- Native mobile applications (the website is mobile-responsive instead).

2.6 Assumptions & Constraints

Assumptions

- The school will appoint at least one staff member as content editor and one as administrator.
- Initial content (history, leadership profiles, photos) will be supplied by the school during onboarding.
- Reliable internet hosting (local or cloud) is available within the project budget.

Constraints

- Staff are non-technical; all routine operations must be achievable through guided forms.
- The interface and all public content must be fully bilingual (Khmer and English).

- The solution must perform acceptably on mobile devices over typical Cambodian mobile networks.
- Khmer typography must render correctly across browsers, requiring a Unicode Khmer web font.

3. Target Users & Personas

The platform serves two broad audiences: public visitors who consume content, and school staff who produce and manage it. The personas below guide design and prioritization decisions throughout the project.

3.1 User Groups

Group	Members	Primary Goals
Public Visitors	Students, parents, community, alumni, NGOs, government officers	Find information quickly; verify the school; read news and achievements
School Staff	Director, Vice Directors, administrative staff	Publish and maintain accurate, timely content without technical help
Website Administrator	Designated IT-capable staff or vendor	Manage users, permissions, configuration, and system integrity

3.2 Personas

Persona 1 — Sophea, the School Director (Primary Admin)

Attribute	Detail
Role	School Director and project sponsor
Technical comfort	Low — uses a smartphone and basic office software
Key needs	Update headline statistics and approve announcements in minutes
Frustrations	Depending on others for simple website changes; outdated information
Success looks like	Logging in, editing the BAC II Grade A number, and seeing it live on the homepage

Persona 2 — Dara, the Administrative Staff (Content Editor)

Attribute	Detail
Role	Office administrator responsible for daily content
Technical comfort	Moderate — comfortable with Facebook and basic forms
Key needs	Quickly post activities with photos; publish exam and holiday notices
Frustrations	Complicated tools; uploading many photos one at a time
Success looks like	Creating an activity, uploading a photo gallery, and publishing in a few minutes

Persona 3 — Lina, the Parent (Public Visitor)

Attribute	Detail
Role	Parent of a prospective Grade 10 student
Device	Mid-range Android phone on mobile data
Key needs	Admission requirements, registration dates, required documents, contact
Frustrations	Slow pages; information only in one language; broken links
Success looks like	Reading enrollment info in Khmer, downloading the form, and contacting the school

Persona 4 — Vichea, the Alumnus (Public Visitor)

Attribute	Detail
Role	Former student, now a university graduate
Device	Laptop and phone
Key needs	Follow school achievements, activities, and competition results
Frustrations	No official place to see school news beyond social media
Success looks like	Browsing recent achievements and the photo gallery, and sharing them

4. Functional Requirements

This section specifies the functional behaviour of the system, organized by module. Each requirement carries a unique identifier (FR-x.y) and a priority: Must (essential for MVP), Should (important but deferrable), or Could (desirable). Acceptance criteria define the verifiable conditions for completion and form the basis of QA test cases.

All public-facing content described below is bilingual: every title and body field exists in both Khmer and English, and visitors switch language through a persistent language toggle.

4.1 Module 1 — Public Website (Home Page)

The home page is the primary entry point and must present a clear, fast, and engaging overview of the school. It aggregates the most important live content from other modules.

Functional Requirements

ID	Requirement	Priority
FR-1.1	Display school introduction and principal's welcome message	Must
FR-1.2	Display a hero statistics strip (students, teachers, classes, BAC II Grade A)	Must
FR-1.3	Display the latest news items pulled automatically from the News module	Must
FR-1.4	Display recent school activities with thumbnails	Must
FR-1.5	Highlight featured achievements	Must
FR-1.6	Provide a quick contact block (address, phone, email, map link)	Must
FR-1.7	Provide a persistent Khmer/English language toggle in the header	Must
FR-1.8	Provide responsive primary navigation to all public modules	Must

Acceptance Criteria

#	Acceptance Criteria
1	The home page is fully responsive on mobile, tablet, and desktop breakpoints.
2	The home page renders meaningful content (Largest Contentful Paint) in under 3 seconds on a 4G connection.
3	Hero statistics reflect the current values stored by the Statistics module without manual code changes.
4	Latest news and activities update automatically when new items are published.
5	Switching language updates all visible labels and content without a full page reload error.

4.2 Module 2 — School Statistics Dashboard

This module lets the Director maintain headline statistics independently. It is the single most important self-service capability in the system because these numbers represent the school's performance publicly.

Editable Statistics

- Total number of students; total number of classes; total number of teachers.
- Number of Grade A students in the BAC II national exam; graduation rate.
- Student gender distribution; school establishment year.

Functional Requirements

ID	Requirement	Priority
FR-2.1	Provide a simple form to update each statistic value	Must
FR-2.2	Tag each statistic value with an academic year	Must
FR-2.3	Record a historical entry every time a value changes	Must
FR-2.4	Display historical statistics as a trend (table and chart)	Should
FR-2.5	Auto-update the homepage hero strip when values change	Must
FR-2.6	Validate numeric ranges and reject invalid input	Must

Acceptance Criteria

#	Acceptance Criteria
1	A Director can update any statistic and save it in under one minute without assistance.
2	Saved changes are visible on the public homepage immediately (no cache delay beyond a few seconds).
3	Each change creates a history record capturing previous value, new value, year, user, and timestamp.
4	Invalid values (e.g. negative counts, rate above 100%) are rejected with a clear bilingual message.

4.3 Module 3 — School Activities Management

Staff publish school activities such as events, sports, and competitions, each with rich media. Activities feed both a dedicated listing page and the home page.

Activity Fields & Categories

- Fields: title, description, activity date, location, photos, videos, category.
- Categories: Academic Activities, Sports, Community Service, Student Clubs, Competitions, Events.

Functional Requirements

ID	Requirement	Priority
FR-3.1	Create, edit, and delete activities	Must
FR-3.2	Publish or unpublish an activity	Must
FR-3.3	Upload multiple images in a single action	Must
FR-3.4	Attach video links or uploads	Should
FR-3.5	Assign an activity to a category	Must
FR-3.6	Display activities in a responsive gallery with filtering by category	Must

Acceptance Criteria

#	Acceptance Criteria
1	An editor can upload several images at once and they appear in the activity gallery.
2	The activity gallery is responsive and displays correctly on mobile and desktop.
3	Unpublished activities are not visible to the public but remain editable in the admin.
4	Category filtering returns only activities within the selected category.

4.4 Module 4 — News & Announcements

This module manages official news and announcements, including time-sensitive notices that can be scheduled in advance.

Announcement Types

- Exam Information, Student Registration, Scholarship Opportunities, Holiday Notices, School Events, Ministry Announcements.

Functional Requirements

ID	Requirement	Priority
FR-4.1	Publish announcements and news articles	Must
FR-4.2	Mark items as featured for prominent display	Must
FR-4.3	Schedule publishing for a future date and time	Should
FR-4.4	Categorize items by announcement type	Must
FR-4.5	Search news by keyword	Must
FR-4.6	Filter news by category	Must

Acceptance Criteria

#	Acceptance Criteria
1	Visitors can search news by keyword and receive relevant results in the selected language.
2	Category filtering correctly narrows the news list.
3	A scheduled item becomes publicly visible automatically at its scheduled time, not before.
4	Featured items are visually highlighted and surfaced on the homepage.

4.5 Module 5 — School Achievement Showcase

Showcases student, teacher, and school achievements, awards, and competition results to build reputation.

Functional Requirements

ID	Requirement	Priority
FR-5.1	Record student, teacher, and school achievements and awards	Must
FR-5.2	Capture title, description, award level, photo, and date	Must
FR-5.3	Record competition results	Should
FR-5.4	Highlight selected achievements on the homepage	Must

Acceptance Criteria

#	Acceptance Criteria
1	Featured achievements appear on the homepage achievement section.
2	Each achievement displays its title, award level, photo, and date.
3	Achievements can be filtered by type (student, teacher, school).

4.6 Module 6 — School Information Management

Manages relatively static institutional content grouped into About, Leadership, and Academic information.

Managed Content

- About School: history, vision, mission, core values.
- Leadership Team: director profile, vice directors, department heads.
- Academic Information: grade levels, programs offered, school calendar.

Functional Requirements

ID	Requirement	Priority
FR-6.1	Edit About content (history, vision, mission, core values)	Must
FR-6.2	Manage leadership profiles with photo, name, position, and bio	Must
FR-6.3	Manage academic information and the school calendar	Must
FR-6.4	Order leadership profiles for display	Should

Acceptance Criteria

#	Acceptance Criteria
1	An editor can update any About content through a rich-text editor and see it reflected publicly.
2	Leadership profiles display in the configured order with photos.
3	Academic information and calendar are presented clearly in both languages.

4.7 Module 7 — Photo & Media Gallery

A central media gallery organizes photo albums and videos by year and category, with featured media for the homepage.

Functional Requirements

ID	Requirement	Priority
FR-7.1	Create photo albums and video collections	Must
FR-7.2	Organize media by year and category	Must
FR-7.3	Mark media as featured	Should
FR-7.4	Display a responsive lightbox gallery view	Must

Acceptance Criteria

#	Acceptance Criteria
1	Galleries are organized and browsable by year and category.
2	Images open in a responsive lightbox on all devices.
3	Featured media can be surfaced on the homepage or relevant pages.

4.8 Module 8 — Contact & Location

Provides school contact details, an embedded map, and a contact form that stores messages and notifies the administrator.

Functional Requirements

ID	Requirement	Priority
FR-8.1	Display address, phone, email, and embedded Google Map	Must
FR-8.2	Provide a contact form (name, email, phone, subject, message)	Must
FR-8.3	Store submitted messages in the system	Must
FR-8.4	Send an email notification to the administrator on submission	Must
FR-8.5	Protect the form against spam (validation + anti-bot measure)	Should

Acceptance Criteria

#	Acceptance Criteria
1	Submitted messages are stored and visible in the admin Messages area.
2	An email notification is delivered to the administrator within one minute of submission.
3	The embedded map displays the correct school location.
4	Form validation prevents empty or malformed submissions with bilingual error messages.

4.9 Recommended MVP Features

Stakeholder analysis confirms four additional features deliver high value for relatively low effort and are included in the MVP.

4.9.1 Student Enrollment Information

Displays admission requirements, registration process, required documents, and important dates. This is one of the most common reasons parents visit a school website, so it is prominent and bilingual.

4.9.2 Download Center

Allows visitors to download registration forms, school policies, the academic calendar, and official documents. This reduces administrative workload by serving common requests automatically. Each document records a download count.

4.9.3 Notice Board

A quick-update area for urgent announcements, exam schedules, and public notices, with an optional urgency flag and start/end display dates.

4.9.4 Hero Statistics Section

A visual dashboard on the homepage showing total students, teachers, classes, and BAC II Grade A students, drawn directly from the Statistics module so school performance is communicated immediately.

4.10 Admin Back Office

The back office is the operational heart of the system. It must be clean, guided, and bilingual so that non-technical staff can work confidently.

4.10.1 Dashboard

- Summary cards: website visitors, published activities, published news, current statistics, recent updates.
- A visitor-trend chart and a recent-activity feed.

4.10.2 Content Management

- Unified management of news, activities, gallery, documents, and school information.
- Draft, publish, unpublish, and (soft) delete states for content.

4.10.3 User Management

Three roles ship with the MVP, each mapped to a set of permissions across content types.

Role	Create	Edit	Delete	Publish	Manage Users
Administrator	Yes	Yes	Yes	Yes	Yes
Director	Yes	Yes	Yes	Yes	No
Content Editor	Yes	Yes	No	No	No

Permissions are enforced on both the interface and the API. Every create, edit, delete, and publish action is written to the audit log (see Non-Functional Requirements).

5. User Stories

User stories express requirements from the perspective of each user role using the standard form: As a [role], I want [capability], so that [benefit]. Each story includes acceptance criteria expressed in Given/When/Then style for QA traceability. Stories are grouped by module and carry the relevant functional-requirement references.

5.1 Director & Statistics

ID	User Story	Acceptance Criteria (summary)
US-01	As a Director, I want to update school statistics through a simple form, so that the public always sees current numbers.	Given I am logged in as Director, when I edit a statistic and save, then the homepage reflects the new value within seconds and a history record is created.
US-02	As a Director, I want to see the history of a statistic, so that I can track progress year over year.	Given a statistic has previous values, when I open its history, then I see prior values with year, user, and timestamp.
US-03	As a Director, I want to approve and publish announcements, so that official information is controlled.	Given a draft announcement, when I publish it, then it becomes publicly visible and is logged.

5.2 Content Editor

ID	User Story	Acceptance Criteria (summary)
US-04	As an Editor, I want to create an activity with multiple photos, so that I can showcase events quickly.	Given the activity form, when I upload several images and save, then they appear in the activity gallery.
US-05	As an Editor, I want to schedule a notice, so that it appears automatically on the right date.	Given a scheduled publish time, when that time arrives, then the notice becomes public automatically.
US-06	As an Editor, I want to write content in Khmer and English, so that all visitors are served.	Given a content form, when I fill both language fields, then visitors see content in their chosen language.
US-07	As an Editor, I want to upload downloadable documents, so that parents can self-serve forms.	Given the Download Center, when I upload a document, then it is listed and downloadable, and downloads are counted.

5.3 Administrator

ID	User Story	Acceptance Criteria (summary)
US-08	As an Administrator, I want to manage users and roles, so that only authorized staff can publish.	Given the user manager, when I assign a role, then that user gains exactly the permissions of that role.
US-09	As an Administrator, I want an audit log of all changes, so that I can trace who changed what.	Given any create/edit/delete/publish action, when it occurs, then an audit record stores user, action, entity, and time.
US-10	As an Administrator, I want to read contact messages, so that enquiries are not missed.	Given a submitted contact form, when I open Messages, then I see the message and can mark it read.

5.4 Public Visitors

ID	User Story	Acceptance Criteria (summary)
US-11	As a Parent, I want to find enrollment requirements and dates, so that I can register my child.	Given the Enrollment page, when I open it, then I see requirements, documents, process, and dates in my language.
US-12	As a Parent, I want to download the registration form, so that I can prepare in advance.	Given the Download Center, when I select the form, then it downloads successfully.
US-13	As a Student, I want to search news for exam information, so that I find it quickly.	Given the news search, when I enter a keyword, then matching items are returned in my language.
US-14	As an Alumnus, I want to browse achievements and the gallery, so that I can follow the school.	Given the achievements and gallery pages, when I browse, then content is organized and shareable.
US-15	As any Visitor, I want to switch between Khmer and English, so that I can read comfortably.	Given the language toggle, when I switch language, then labels and content update accordingly.
US-16	As any Visitor, I want the site to load fast on my phone, so that I do not abandon it.	Given a mobile 4G connection, when I open the homepage, then it loads in under 3 seconds.

6. Non-Functional Requirements

Non-functional requirements (NFRs) define the quality attributes the system must satisfy. They are as important as functional requirements and are testable. Each NFR carries an identifier and a measurable target.

6.1 Performance

ID	Requirement	Target
NFR-P1	Homepage load time on mobile (4G)	Under 3 seconds (LCP)
NFR-P2	Interior page load time	Under 3 seconds
NFR-P3	Statistic update reflected publicly	Within a few seconds
NFR-P4	Image delivery	Responsive, compressed, lazy-loaded
NFR-P5	Concurrent visitors supported (MVP)	At least 200 concurrent without degradation

6.2 Security

- Secure authentication with hashed passwords (bcrypt or Argon2) and protection against brute-force login.
- Role-based access control (RBAC) enforced on both the user interface and every API endpoint.
- Audit logs recording user, action, affected entity, IP address, and timestamp for all content changes.
- Protection against the common OWASP Top 10 risks: SQL injection, XSS, CSRF, and insecure direct object references.
- All traffic served over HTTPS/TLS; secure, HTTP-only cookies for refresh tokens.
- File-upload validation (type, size, and content) to prevent malicious uploads.
- Sensitive data and credentials stored in environment configuration, never in source code.

6.3 Usability & Accessibility

- Full Khmer and English support across the public site and admin panel, with a persistent language toggle.
- Correct Khmer Unicode rendering using a bundled Khmer web font (e.g. a Noto/Khmer OS family).
- A user-friendly admin panel designed for non-technical staff, with guided forms and clear bilingual labels.
- Mobile-first, responsive layouts across mobile, tablet, and desktop breakpoints.
- Adherence to WCAG 2.1 AA where practical: sufficient color contrast, keyboard navigation, and alt text on images.

6.4 SEO & Discoverability

- Server-rendered or pre-rendered public pages with semantic, search-engine-friendly markup.
- Configurable meta titles and descriptions per page, with localized variants.
- Open Graph and Twitter Card tags so shared links present with image, title, and description.
- Automatic XML sitemap generation and a robots.txt; clean, human-readable URL slugs.
- Structured data (schema.org) for the organization and news articles where appropriate.

6.5 Reliability, Maintainability & Compatibility

ID	Requirement	Target / Note
NFR-R1	Availability	99.5% monthly uptime target
NFR-R2	Backups	Daily automated database and media backups
NFR-R3	Browser support	Latest Chrome, Firefox, Safari, Edge; Android Chrome; iOS Safari
NFR-R4	Maintainability	Modular codebase, documented, version-controlled (Git)
NFR-R5	Localization	All UI strings externalized for translation
NFR-R6	Scalability	Stateless API able to scale horizontally behind a load balancer

7. System Architecture & Technology Stack

7.1 Architecture Overview

The system follows a decoupled architecture: a single-page front-end application consumes a RESTful API served by the back-end, which in turn persists data in a relational database and stores uploaded media on local or cloud storage. This separation keeps the public site fast, allows the admin panel to reuse the same API, and makes future integrations (mobile apps, portals) straightforward.

The public site should be server-side rendered or statically pre-rendered for SEO and first-load performance, while the admin back office runs as an authenticated single-page application. A caching layer in front of read-heavy public endpoints keeps the homepage fast even under load.

Logical Tiers

Tier	Responsibility	Technology
Presentation	Public website + admin SPA, bilingual UI, responsive layouts	Vue.js 3 (Composition API), Pinia, Tailwind CSS
Application / API	Business logic, validation, authentication, RBAC, REST API	Laravel 12 (PHP) — or Node.js/Express alternative
Data	Relational persistence, relationships, transactions	MySQL 8
Storage	Images, videos, documents	Local storage (MVP) or cloud object storage
Cross-cutting	Caching, queues (email), logging, backups	Cache + queue worker; email service

7.2 Technology Stack

Frontend

- Vue.js 3 with the Composition API for a maintainable, component-based UI.
- Pinia for predictable state management (language, auth session, content stores).
- Tailwind CSS for a consistent, responsive design system and rapid styling.
- Vue Router for navigation; Vue I18n for Khmer/English localization.

Backend

Two stacks satisfy the requirements; the recommendation and rationale follow.

Option	Strengths	Considerations
Laravel 12 (PHP) — recommended	Batteries-included: auth, validation, ORM (Eloquent), queues, migrations; large local developer talent pool; excellent admin tooling	PHP hosting required (widely available and affordable in Cambodia)
Node.js + Express	JavaScript end-to-end; lightweight; large ecosystem	More boilerplate for auth, validation, and ORM; team must assemble conventions

Recommendation: Laravel 12 with MySQL is recommended for the MVP. Its built-in authentication, validation, Eloquent ORM, migrations, and queue system map directly onto this project's needs and shorten delivery time, while affordable PHP/MySQL hosting and a strong local talent pool support long-term sustainability. The architecture remains valid if the team instead selects Node.js/Express.

Database & Storage

- MySQL 8 as the primary relational database, accessed through the ORM with versioned migrations.
- Media stored on local disk for the MVP, with a storage abstraction so a move to cloud object storage requires no application changes.

Infrastructure & DevOps

- Git version control with a branch-and-pull-request workflow.
- Separate environments: local, staging, and production, configured via environment variables.
- Automated daily backups of database and media; HTTPS via a managed certificate.
- A queue worker for asynchronous email notifications (contact form).

7.3 High-Level Component View

Component	Description
Public Web App	SSR/pre-rendered Vue site consuming public read API endpoints
Admin SPA	Authenticated Vue application for content and user management
REST API (v1)	Versioned API enforcing validation, RBAC, and audit logging
Auth Service	Login, token issuance/refresh, password hashing
Media Service	Upload handling, image resizing, storage abstraction
Notification Worker	Queued email notifications to administrators
Database	MySQL with the schema defined in Section 8

8. Database Design

The data model is normalized to third normal form and designed for bilingual content: text fields that appear publicly are stored in paired Khmer and English columns (suffixes `_km` and `_en`). The model supports content management, statistics with history, polymorphic media, role-based access control, and audit logging.

8.1 Entity Relationship Diagram (ERD)

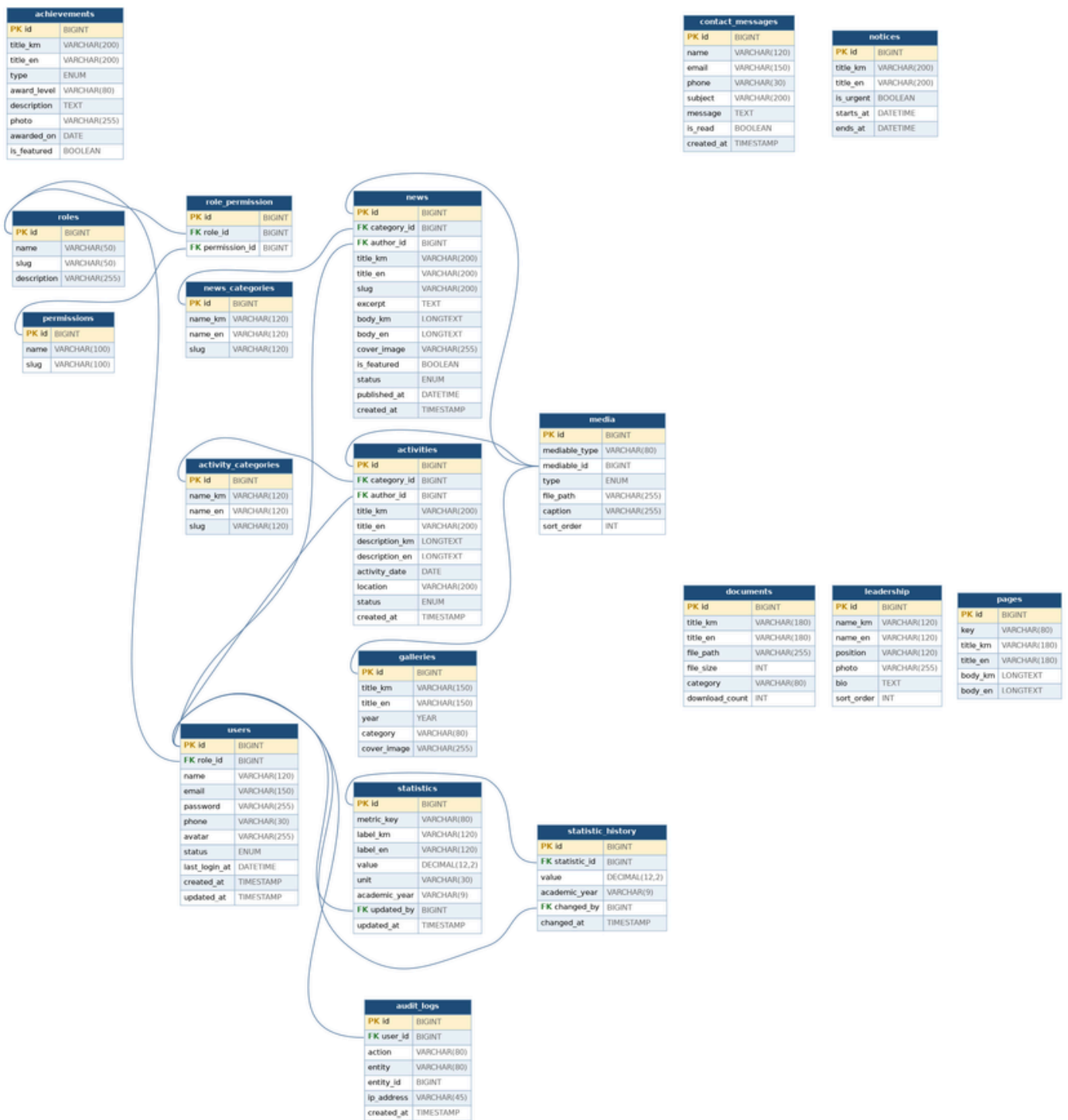


Figure 1 — Entity Relationship Diagram of the High School Website Management System

8.2 Database Schema

The principal tables and their key columns are summarized below. All tables include `created_at` and `updated_at` timestamps unless otherwise noted; primary keys are auto-incrementing BIGINT identifiers.

8.2.1 Identity & Access

Table	Key Columns	Notes
roles	id, name, slug, description	Administrator, Director, Content Editor
permissions	id, name, slug	Granular abilities (create, edit, delete, publish, manage_users)
role_permission	role_id (FK), permission_id (FK)	Many-to-many pivot
users	id, role_id (FK), name, email, password, status, last_login_at	Staff accounts; password hashed
audit_logs	id, user_id (FK), action, entity, entity_id, ip_address, created_at	Records all content changes

8.2.2 Content

Table	Key Columns	Notes
news	id, category_id (FK), author_id (FK), title_km/_en, slug, body_km/_en, cover_image, is_featured, status, published_at	Supports featured + scheduled publishing
news_categories	id, name_km/_en, slug	Announcement types
activities	id, category_id (FK), author_id (FK), title_km/_en, description_km/_en, activity_date, location, status	School activities
activity_categories	id, name_km/_en, slug	Sports, Events, etc.
achievements	id, title_km/_en, type, award_level, description, photo, awarded_on, is_featured	Student/teacher/school
pages	id, key, title_km/_en, body_km/_en	About, vision, mission, academic info
leadership	id, name_km/_en, position, photo, bio, sort_order	Director, vice directors, heads
documents	id, title_km/_en, file_path, file_size, category, download_count	Download Center
notices	id, title_km/_en, is_urgent, starts_at, ends_at	Notice Board

8.2.3 Media, Statistics & Communication

Table	Key Columns	Notes
media	id, mediable_type, mediable_id, type, file_path, caption, sort_order	Polymorphic: attaches to news, activities, galleries
galleries	id, title_km/_en, year, category, cover_image	Albums organized by year/category
statistics	id, metric_key, label_km/_en, value, unit, academic_year, updated_by (FK)	Headline statistics
statistic_history	id, statistic_id (FK), value, academic_year, changed_by (FK), changed_at	Historical tracking
contact_messages	id, name, email, phone, subject, message, is_read, created_at	Contact form submissions

8.3 Key Table Relationships

- users → roles: many users belong to one role (many-to-one).
- roles ↔ permissions: many-to-many through role_permission.
- news → news_categories and news → users (author): many-to-one each.
- activities → activity_categories and activities → users (author): many-to-one each.
- media → (news | activities | galleries): polymorphic many-to-one via mediable_type / mediable_id.
- statistics → statistic_history: one-to-many (each value change creates a history row).
- users → audit_logs: one-to-many (each user generates many audit entries).

8.4 Indexing Strategy

- Unique indexes on users.email, and on slug columns for news, activities, and categories.
- Composite index on (status, published_at) for news and activities to speed public listing queries.
- Index on media (mediable_type, mediable_id) for fast attachment lookups.
- Index on statistics.metric_key and statistic_history.statistic_id for dashboard reads.

8.5 Migration Plan

The schema is created and evolved through versioned migrations checked into source control, so every environment can be rebuilt deterministically. Seeders populate baseline data.

Step	Migration / Seed	Purpose
M1	Create roles, permissions, role_permission, users	Identity and access foundation
M2	Seed default roles and an initial administrator account	First login and RBAC baseline
M3	Create content tables (news, activities, categories, pages, leadership)	Core content management
M4	Create achievements, documents, notices	Showcase and self-service features
M5	Create media and galleries	Polymorphic media support
M6	Create statistics and statistic_history	Statistics dashboard
M7	Create contact_messages and audit_logs	Communication and traceability
M8	Seed sample categories and demo content (staging only)	Enable testing and UAT

Migrations are forward-only in production and always reversible in development. Destructive changes are introduced through additive steps (add new column, backfill, switch, remove) to avoid data loss.

9. API Specifications

The back end exposes a versioned RESTful API under the `/api/v1` prefix. Public read endpoints are unauthenticated and cacheable; all write endpoints and the entire admin surface require a valid access token and the appropriate role permission. Requests and responses use JSON. List endpoints support pagination, search, and filtering through query parameters.

9.1 Conventions

- Base URL: `https://{domain}/api/v1`
- Authentication: Bearer access token in the Authorization header (see 9.4).
- Localization: an Accept-Language header (km or en) or a `?lang=` query parameter selects the response language for resolved labels.
- Pagination: `?page` and `?per_page`; responses include a meta object with totals.
- Errors: standard HTTP status codes with a consistent JSON error envelope.

9.2 Core Endpoints

9.2.1 Public (read-only)

Method	Endpoint	Purpose
GET	/news	List published news (search, category, pagination)
GET	/news/{slug}	Retrieve a single news article
GET	/activities	List published activities (filter by category)
GET	/activities/{id}	Retrieve a single activity with media
GET	/achievements	List achievements (filter by type)
GET	/statistics	Current headline statistics for the homepage
GET	/galleries	List galleries by year/category
GET	/documents	List downloadable documents
GET	/notices	Active notices for the notice board
GET	/pages/{key}	Static page content (about, vision, academic)
POST	/contact	Submit a contact message (rate-limited)

9.2.2 Authenticated (admin)

Method	Endpoint	Purpose / Permission
POST	/auth/login	Authenticate, issue tokens
POST	/auth/refresh	Rotate access token
POST	/auth/logout	Invalidate refresh token
GET	/admin/dashboard	Dashboard summary metrics
POST/PUT/DELETE	/admin/news/{id?}	Manage news (create/edit/delete/publish)
POST/PUT/DELETE	/admin/activities/{id?}	Manage activities + media upload
PUT	/admin/statistics/{id}	Update a statistic (Director/Admin)
GET	/admin/statistics/{id}/history	Statistic history
POST/PUT/DELETE	/admin/achievements/{id?}	Manage achievements
POST/PUT/DELETE	/admin/documents/{id?}	Manage downloadable documents
GET/PUT	/admin/messages/{id?}	Read / mark contact messages
GET/POST/PUT/DELETE	/admin/users/{id?}	User management (Admin only)
GET	/admin/audit-logs	View audit trail (Admin only)

9.3 Request / Response Samples

9.3.1 Get current statistics

Request

```
GET /api/v1/statistics?lang=en

Accept: application/json
```

Response

```
200 OK

{
  "data": [
    { "metric_key": "total_students", "label": "Total Students",
      "value": 1850, "unit": "students", "academic_year": "2025-2026" },
    { "metric_key": "bac2_grade_a", "label": "BAC II Grade A",
      "value": 76, "unit": "students", "academic_year": "2025-2026" }
  ],
  "meta": { "updated_at": "2026-06-09T08:15:00Z" }
}
```

9.3.2 Update a statistic

Request

```
PUT /api/v1/admin/statistics/12

Authorization: Bearer <access_token>

Content-Type: application/json

{ "value": 80, "academic_year": "2025-2026" }
```

Response

200 OK

```
{  
  
  "data": { "id": 12, "metric_key": "bac2_grade_a",  
  
            "value": 80, "academic_year": "2025-2026" },  
  
  "message": "Statistic updated and history recorded."  
}
```

9.3.3 Submit a contact message

Request

POST /api/v1/contact

Content-Type: application/json

```
{ "name": "Lina", "email": "lina@example.com",  
  
  "phone": "0XXXXXXXX", "subject": "Enrollment",  
  
  "message": "What documents are required for Grade 10?" }
```

Response

201 Created

```
{ "message": "Your message has been received. Thank you." }
```

9.3.4 Error envelope

Response

422 Unprocessable Entity

```
{  
  
  "error": "validation_failed",  
  
  "messages": { "email": ["The email field is invalid."] }  
  
}
```

9.4 Authentication Flow

Authentication uses short-lived JSON Web Token (JWT) access tokens with longer-lived refresh tokens stored in secure, HTTP-only cookies. Every authenticated request is checked by middleware that verifies the token and confirms the user's role holds the required permission (RBAC). Authorized write actions are recorded in the audit log.

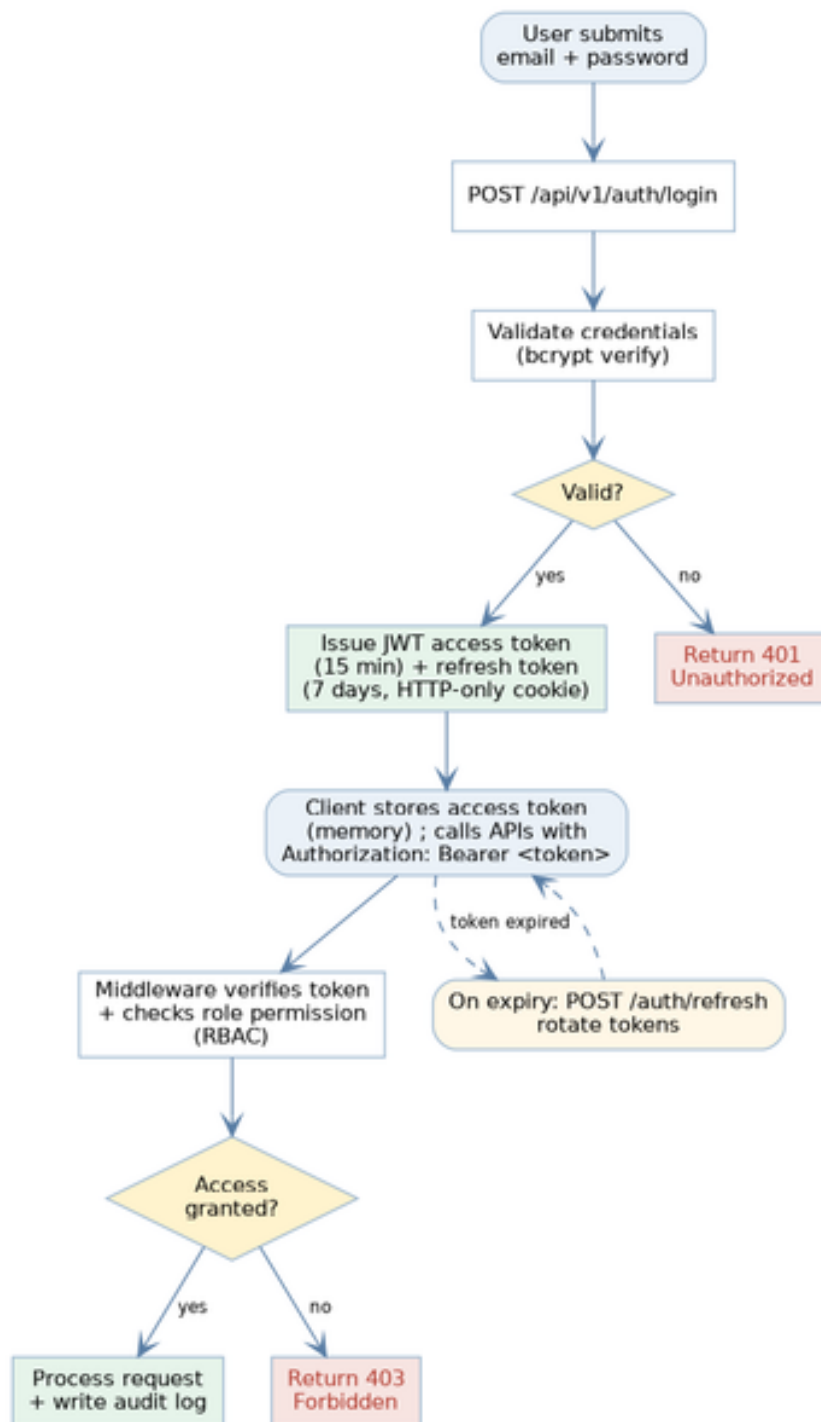


Figure 2 — Authentication and authorization flow (JWT + RBAC)

Status	Meaning
200 / 201	Success / resource created
400	Malformed request
401	Missing or invalid authentication
403	Authenticated but lacking permission
404	Resource not found
422	Validation failed
429	Rate limit exceeded (e.g. contact form)

10. UI/UX Requirements

The interface design prioritizes clarity, speed, and bilingual accessibility. The public site must feel modern and trustworthy; the admin panel must feel simple and guided for non-technical staff. This section defines the sitemap, key user flows, and wireframes for the homepage, admin dashboard, and mobile experience.

10.1 Design Principles

- Mobile-first: design for small screens first, then enhance for larger viewports.
- Bilingual by default: every screen supports a Khmer/English toggle and correct Khmer typography.
- Clarity over decoration: generous spacing, clear hierarchy, and legible Khmer and Latin fonts.
- Consistency: a shared component library (buttons, cards, forms) across public and admin.
- Accessibility: sufficient contrast, focus states, keyboard support, and descriptive alt text.

10.2 Sitemap

The sitemap below shows the public website and the admin back office. Public detail pages (news, activity, album) sit beneath their listing pages.

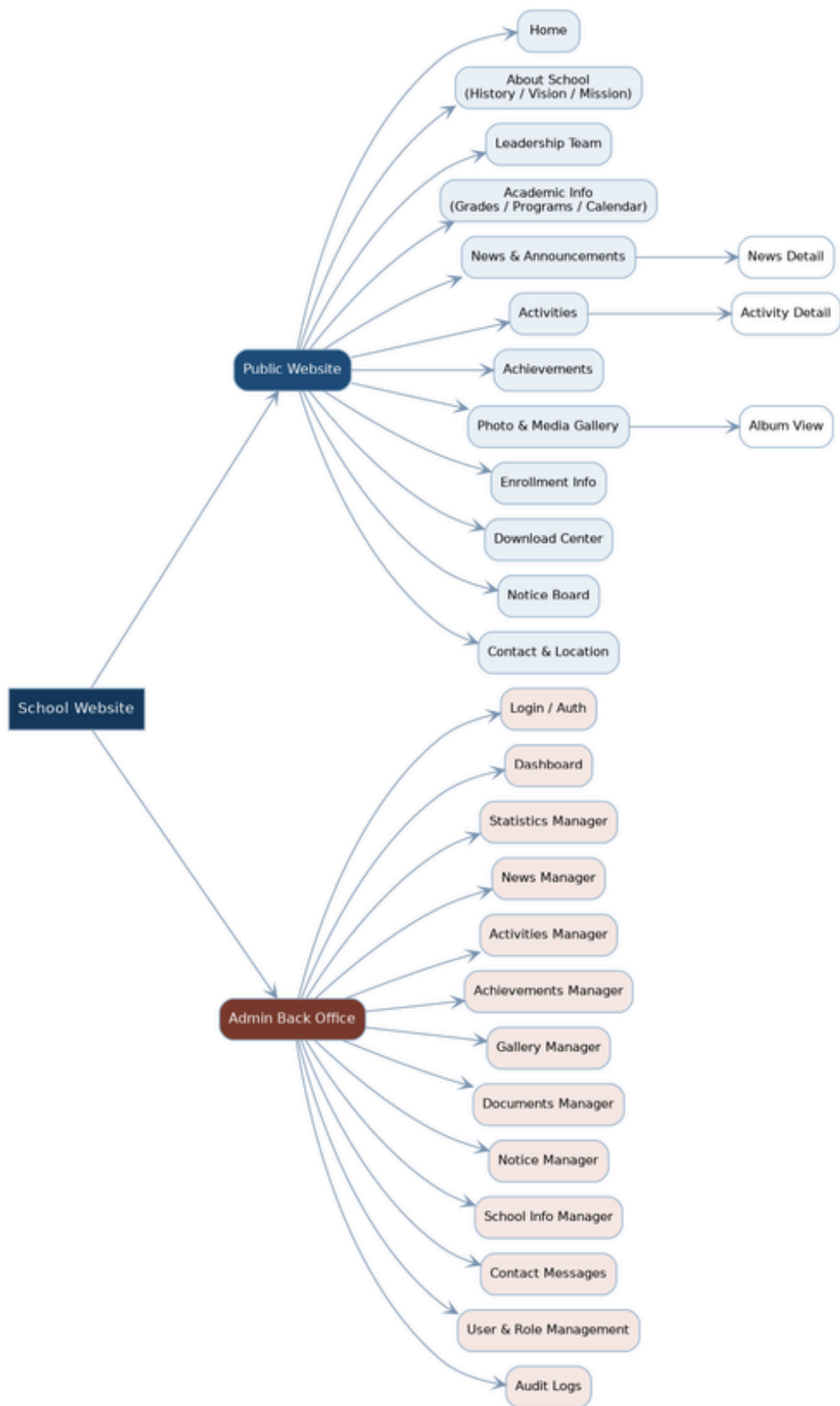


Figure 3 — Site map of public website and admin back office

10.3 Key User Flows

Two representative flows illustrate the most important journeys: a Director updating headline statistics, and a parent finding enrollment information and contacting the school.

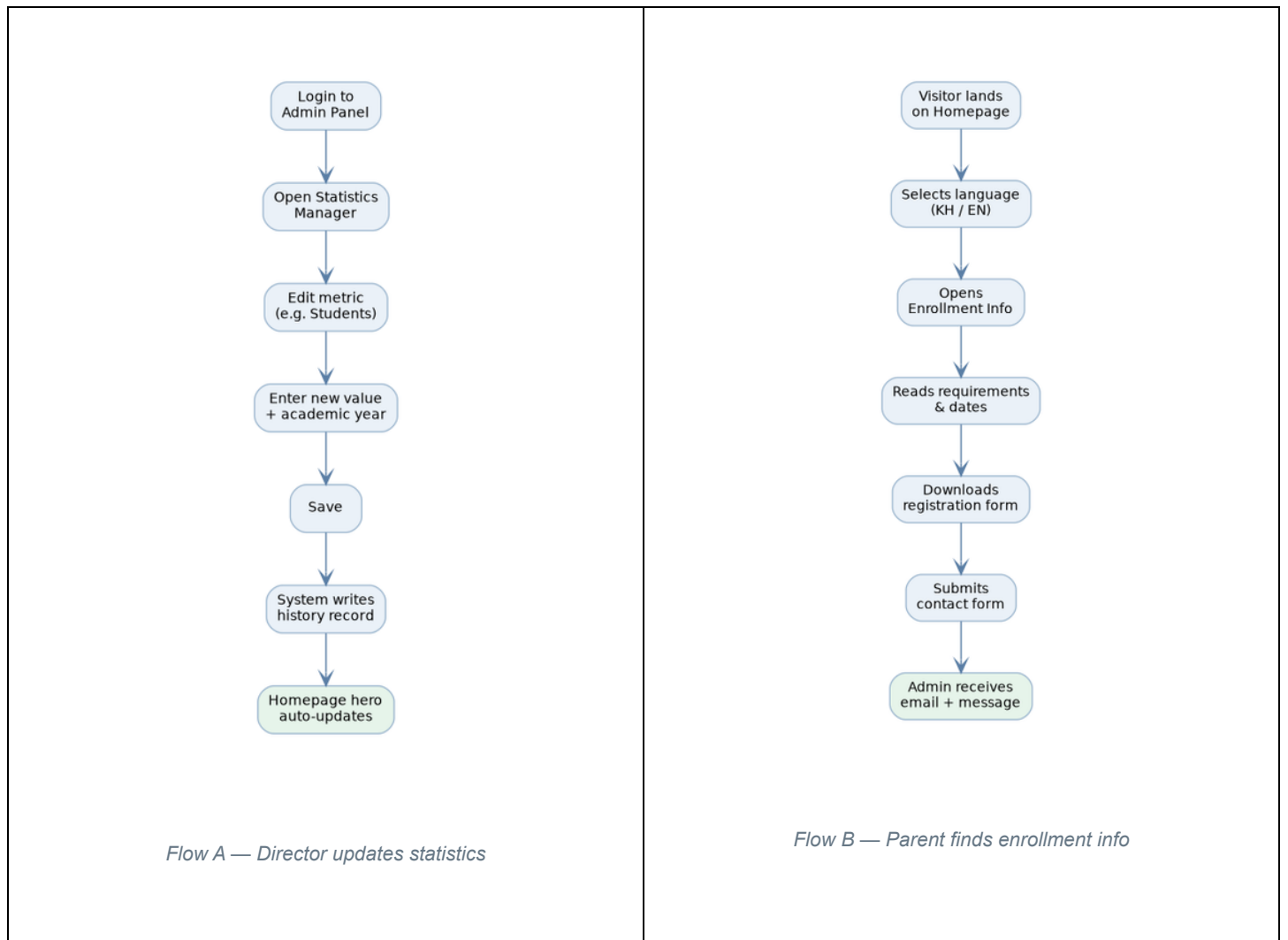


Figure 4 — Representative user flows

10.4 Homepage Layout (Desktop)

The homepage is composed of stacked sections: header with navigation and language toggle, a hero banner, the hero statistics strip, the principal's welcome, latest news, recent activities, featured achievements, an about/quick-contact band, and a footer.

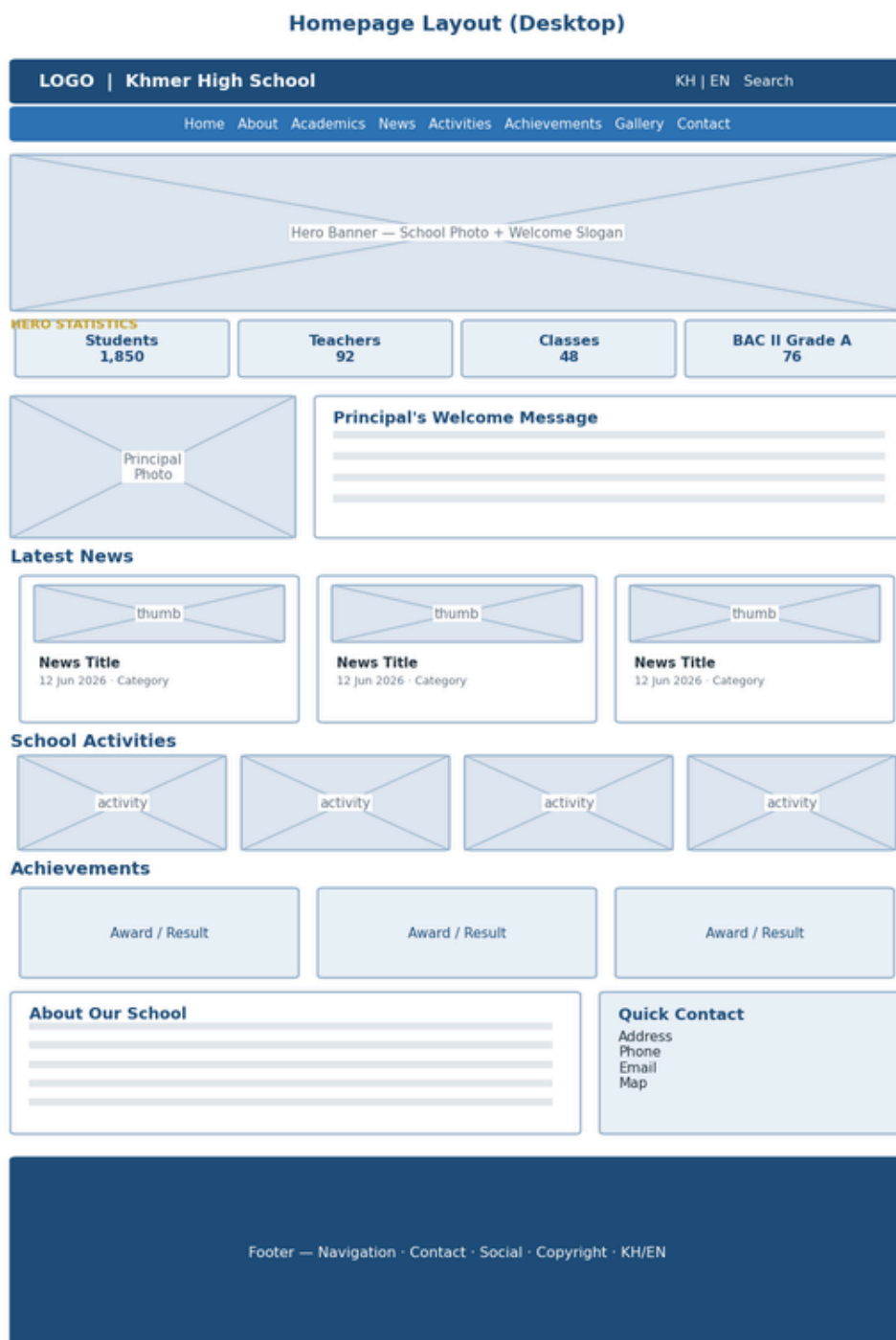


Figure 5 — Homepage wireframe (desktop)

10.5 Mobile Design

On mobile, navigation collapses into a hamburger menu, the hero statistics reflow into a two-by-two grid, and content sections stack vertically with touch-friendly cards. All interactive targets meet minimum touch-size guidance.



Figure 6 — Mobile homepage wireframe

11. Admin Dashboard Design

The admin back office uses a fixed left navigation, a top bar with the user menu and notifications, and a content area. The default landing screen is a dashboard that summarizes site health and recent activity so administrators immediately understand the state of the website.

11.1 Dashboard Layout

Admin Dashboard Layout

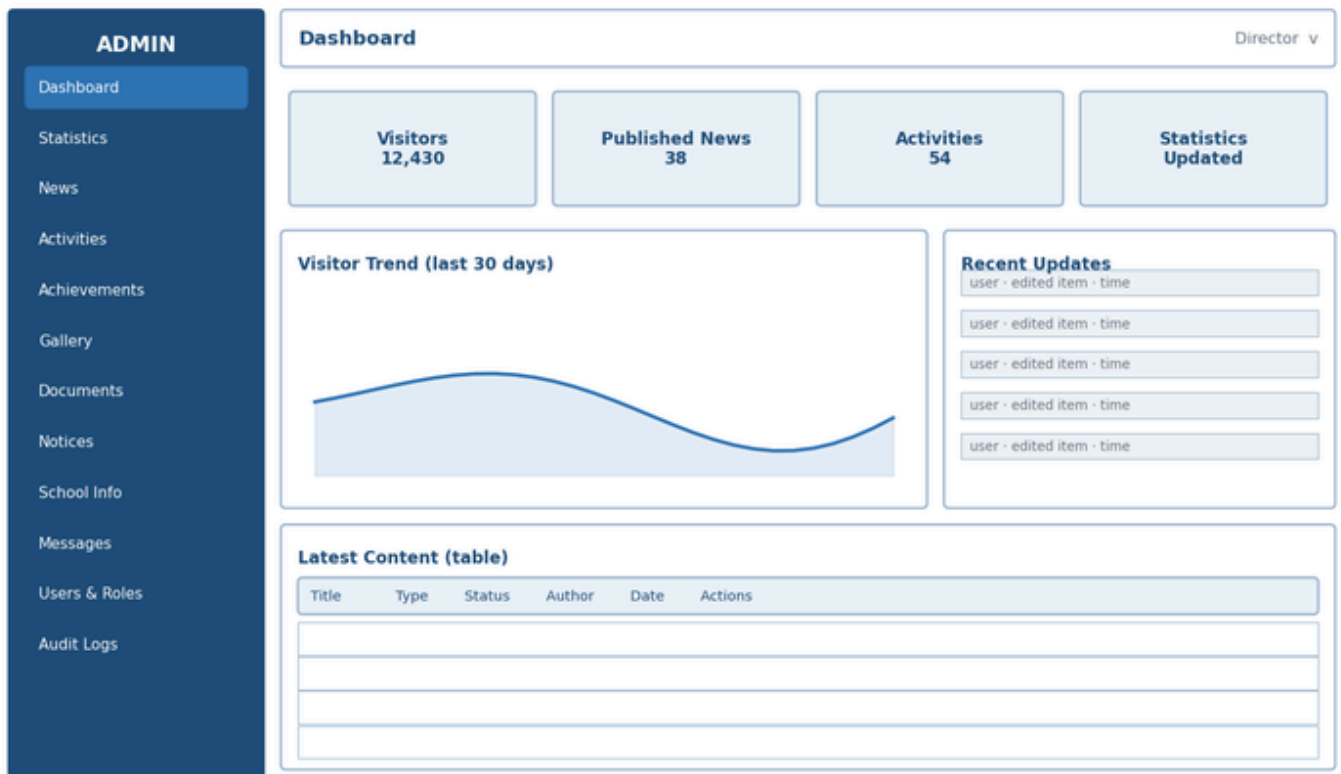


Figure 7 — Admin dashboard wireframe

11.2 Dashboard Components

Component	Content
KPI cards	Website visitors, published news, published activities, statistics status
Visitor trend chart	Visitor counts over the last 30 days
Recent updates feed	Latest content changes with user and time
Latest content table	Recent items with title, type, status, author, date, and actions
Left navigation	Dashboard, Statistics, News, Activities, Achievements, Gallery, Documents, Notices, School Info, Messages, Users & Roles, Audit Logs
Top bar	Language toggle, notifications, current user menu

11.3 Interaction & Content States

- Forms provide inline validation with bilingual error messages and disable submit until valid.
- Content items move through Draft, Published, and Unpublished states; deletion is a soft delete.
- Bulk image upload shows per-file progress and supports reordering before save.
- Destructive actions require a confirmation dialog and are written to the audit log.
- The language toggle persists across the session for the editing experience.

12. Development Roadmap

The MVP is delivered in five sprints over an indicative twelve-week schedule, followed by user acceptance testing and launch. The roadmap is iterative: each sprint produces a demonstrable increment, and the public homepage is wired to live data as early as possible so stakeholders can give feedback continuously.

12.1 Phased Plan

Phase	Duration	Key Deliverables
0 — Setup & Design	Weeks 1–2	Environment setup, CI, design system, finalize wireframes, database migrations (M1–M2), authentication
1 — Core Content	Weeks 3–5	News, activities, categories, school information, leadership; admin shell and RBAC
2 — Statistics & Media	Weeks 5–7	Statistics dashboard with history and homepage hero; media galleries; achievements
3 — Self-service & Contact	Weeks 7–9	Enrollment info, download center, notice board, contact form + email notifications
4 — Public Site & SEO	Weeks 9–11	Public homepage and all listing/detail pages, bilingual toggle, SEO, Open Graph, sitemap
5 — Hardening & UAT	Weeks 11–12	Performance tuning, security review, accessibility, UAT, content load, launch

12.2 Indicative Timeline

Week	1	2	3	4	5	6	7	8	9	10	11	12
Setup & Design	X	X										
Core Content			X	X	X							
Statistics & Media					X	X	X					
Self-service & Contact							X	X	X			
Public Site & SEO									X	X	X	
Hardening & UAT											X	X

Durations are indicative and assume a small team (typically one backend developer, one frontend developer, a part-time designer, and a part-time QA engineer). Sprint boundaries overlap intentionally to keep work flowing.

12.3 Team & Responsibilities

Role	Responsibility
Project Manager	Planning, stakeholder communication, scope and risk management
Solution Architect	Architecture, data model, API contracts, technical standards
Backend Developer	API, business logic, RBAC, migrations, integrations
Frontend Developer	Public site and admin SPA, responsive bilingual UI
UI/UX Designer	Design system, wireframes, visual design, usability
QA Engineer	Test cases from acceptance criteria, functional and regression testing

13. MVP Scope

The MVP delivers a complete, production-ready public website and administrative back office covering every module and recommended feature defined in this document. The table below makes the boundary explicit so expectations are aligned before development begins.

13.1 Included in MVP

Area	Included
Public modules	Home, About/Leadership/Academic, News & Announcements, Activities, Achievements, Gallery, Contact & Location
Recommended features	Enrollment information, Download Center, Notice Board, Hero Statistics
Statistics	Self-service dashboard with history and automatic homepage update
Administration	CMS for all content, media management, users & roles (3 roles), audit logs, dashboard
Languages	Full Khmer and English across public site and admin
Integrations	Google Maps embed, email notifications for contact messages
Quality	Responsive design, performance (<3s), security (RBAC, HTTPS), SEO + Open Graph + sitemap

13.2 Explicitly Excluded from MVP

- Student/parent login accounts, grade portals, attendance, or fee payment.
- E-learning, online classes, quizzes, or digital library.
- SMS or Telegram notifications and social-media auto-posting.
- Multi-campus management and content-approval workflows.
- Native iOS/Android applications.

Excluded items are not rejected — several are scheduled for Phase 2 (Section 14). Keeping them out of the MVP protects the timeline and budget and ensures the essential public presence and self-service capability ship first.

14. Future Phase 2 Features

Once the MVP is live and the school is comfortably maintaining it, the following enhancements add depth and reach. They are listed with an indicative value/effort assessment to support prioritization.

Feature	Description	Value	Effort
Content approval workflow	Draft → review → approve → publish with notifications	High	Medium
Parent / student portal	Logins for grades, schedules, and personalized notices	High	High
SMS & Telegram notifications	Push urgent notices to parents via popular channels	High	Medium
Online enrollment / application	Submit applications and documents online	High	High
Analytics suite	Built-in dashboards for traffic and content performance	Medium	Medium
Multi-campus support	Manage several campuses from one back office	Medium	High
Alumni network	Alumni directory, registration, and updates	Medium	Medium
Event calendar & RSVP	Interactive calendar with event registration	Medium	Medium
E-library / resources	Downloadable learning resources by subject and grade	Medium	Medium
Newsletter subscriptions	Email subscriptions for news and notices	Low	Low
Live chat / chatbot	Bilingual assistant for common enrollment questions	Medium	High

A natural Phase 2 sequence prioritizes the content-approval workflow and SMS/Telegram notifications first (high value, moderate effort), followed by the parent/student portal and online enrollment as larger investments.

15. Acceptance Criteria & Quality Assurance

Quality is verified against the acceptance criteria embedded in Section 4 and the non-functional targets in Section 6. Each acceptance criterion becomes one or more test cases, giving full traceability from requirement to test.

15.1 Testing Strategy

Test Type	Scope	Owner
Unit tests	Business logic, validation, permission checks	Developers
Integration tests	API endpoints, database, authentication, RBAC	Developers / QA
Functional tests	Each module against its acceptance criteria	QA
Responsive / cross-browser	Layouts on target browsers and devices	QA
Performance tests	Load time and concurrency targets (Section 6.1)	QA
Security review	OWASP Top 10, auth, file uploads, access control	Architect / QA
Accessibility checks	Contrast, keyboard, alt text (WCAG 2.1 AA)	QA / Designer
User acceptance testing	Stakeholder validation against real workflows	School staff

15.2 Global Acceptance Criteria

In addition to per-module criteria, the system as a whole must satisfy the following before launch:

- All public pages are fully available and consistent in both Khmer and English.
- The homepage and interior pages meet the under-three-second load target on mobile.
- Non-technical staff complete the core tasks (update a statistic, publish news, post an activity) unaided during UAT.
- RBAC correctly prevents each role from performing actions outside its permissions.
- Every content change is captured in the audit log.
- Contact submissions are stored and trigger an email notification within one minute.
- SEO essentials are present: meta tags, Open Graph, sitemap.xml, and clean URLs.

15.3 Definition of Done

- Code reviewed, merged, and covered by passing automated tests.
- Acceptance criteria for the item verified by QA.
- Bilingual content and labels present and correct.
- Responsive and accessible on target devices; no critical or high-severity defects open.
- Documentation and any new configuration recorded.

16. Appendix

16.1 Glossary

Term	Definition
BAC II	The Cambodian national high-school exit examination (Baccalaureate II)
CMS	Content Management System — the admin back office for managing content
RBAC	Role-Based Access Control — permissions granted by role
JWT	JSON Web Token — a signed token used for stateless authentication
SPA	Single-Page Application — a web app that updates content without full reloads
SSR	Server-Side Rendering — rendering pages on the server for speed and SEO
LCP	Largest Contentful Paint — a metric for perceived load time
Open Graph	Metadata standard controlling how links appear when shared on social media
ERD	Entity Relationship Diagram — a visual model of database tables and relationships
MVP	Minimum Viable Product — the smallest release that delivers core value
UAT	User Acceptance Testing — validation by end users before launch

16.2 Requirement Traceability (Summary)

Each functional requirement (FR-x.y) traces to one or more user stories (US-nn) and acceptance criteria, and onward to QA test cases. The mapping below shows the primary links; the full matrix is maintained in the project's test-management tool.

Module	FR Range	User Stories	Verified By
Home Page	FR-1.x	US-15, US-16	Functional, performance, responsive
Statistics	FR-2.x	US-01, US-02	Functional, integration
Activities	FR-3.x	US-04	Functional, responsive
News & Announcements	FR-4.x	US-03, US-05, US-13	Functional
Achievements	FR-5.x	US-14	Functional
School Information	FR-6.x	US-06	Functional
Gallery	FR-7.x	US-14	Functional, responsive
Contact	FR-8.x	US-10	Functional, integration
Enrollment / Downloads	FR-4.9	US-07, US-11, US-12	Functional
Administration	FR-4.10	US-08, US-09	Integration, security

End of Software Requirements Specification — Version 1.0